

MCMC Proof of Correctness

Derek Aguiar
University of Connecticut -- CSE5825

Why does MH work?

- We are using samples from a Markov chain to approximate an intractable distribution
- We have to show that the Markov Chain converges to sample from the intractable distribution we are approximating

Markov chains

- Recall Markov chains, sequence of random variables with

$$P(x^{(n)} = x \mid x^{(1)}, \dots, x^{(n-1)}) = P(x^{(n)} = x \mid x^{(n-1)})$$

- Transition probability $P(x^{(n)} = x \mid x^{(n-1)})$ is referred to as the *transition kernel*
- x 's can be vectors of random variables
- We assume a homogeneous Markov chain, i.e., the transition kernel $P(x^{(t)} = x \mid x^{(t-1)})$ is fixed (denoted $T(x' \mid x)$)



Markov chain properties

- In graphical modelling, the graph represented a family of probability distributions.
- Let $\pi^{(t)}(x)$ be the probability distribution at time t over the state of the Markov chain x
- States transition from $x^{(t)}$ to $x^{(t+1)}$ given $T(x'|x)$, distributions transition from $\pi^{(t)}(x)$ to $\pi^{(t+1)}(x)$ with

$$\pi^{(t+1)}(x') = \sum_x \pi^{(t)}(x) T(x'|x)$$

x' has $\uparrow$ probability $\max_x \pi^{(t+1)}(x')$ at time $t+1$



Markov chain properties: stationarity

- **Stationary distribution:** $\pi(x)$ is stationary if it does not change under the transition kernel:

$$\pi(x') = \sum_x \pi(x) T(x' | x) \quad \text{for all } x'$$

- In the limit, this describes the proportion of time the chain spends in state x'
- We want the MCMC methods described previously to draw samples from a stationary $P(x)$

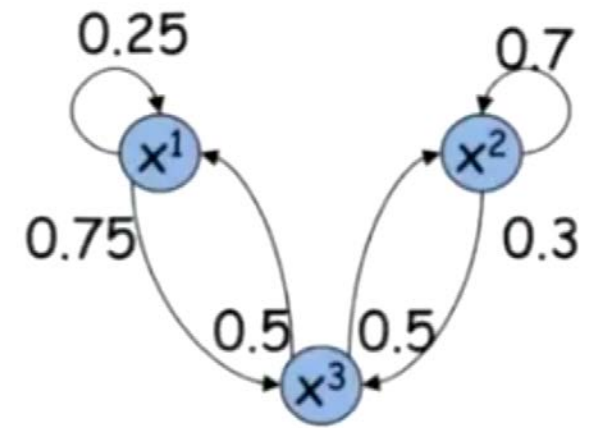
Stationarity example

- Simultaneous equations with $\pi(x^1) + \pi(x^2) + \pi(x^3) = 1$

$$\pi(x^1) = 0.25\pi(x^1) + 0.5\pi(x^3)$$

$$\pi(x^2) = 0.7\pi(x^2) + 0.5\pi(x^3)$$

$$\pi(x^3) = 0.75\pi(x^1) + 0.3\pi(x^2)$$



- Stationary distribution

$$\pi(x^1) = 0.2$$

$$\pi(x^2) = 0.5$$

$$\pi(x^3) = 0.3$$



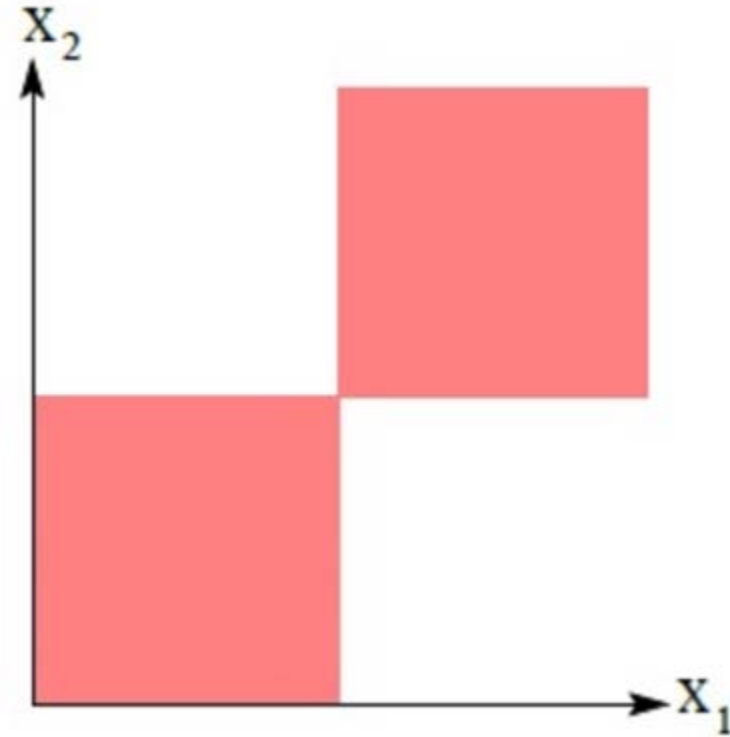
Markov chain properties

- **Irreducibility:** an MC is irreducible if the probability of reaching a state x' from x in a finite number of steps is > 0 for all (x, x')
 - That is, no state is unreachable which is a function of $T(x'|x)$
- **Aperiodicity:** an MC is aperiodic if the chain can revisit state x at any time
 - There are no states that require a specific number of steps to revisit.
- **Ergodic or regular:** an MC is ergodic if it is **irreducible and aperiodic**
- **Theorem:** A regular Markov chain converges to a unique stationary distribution
 - **Corollary:** the stationary distribution $\pi_{st}(x)$ is reachable regardless of the initial distribution



Markov chain for Gibbs Sampling

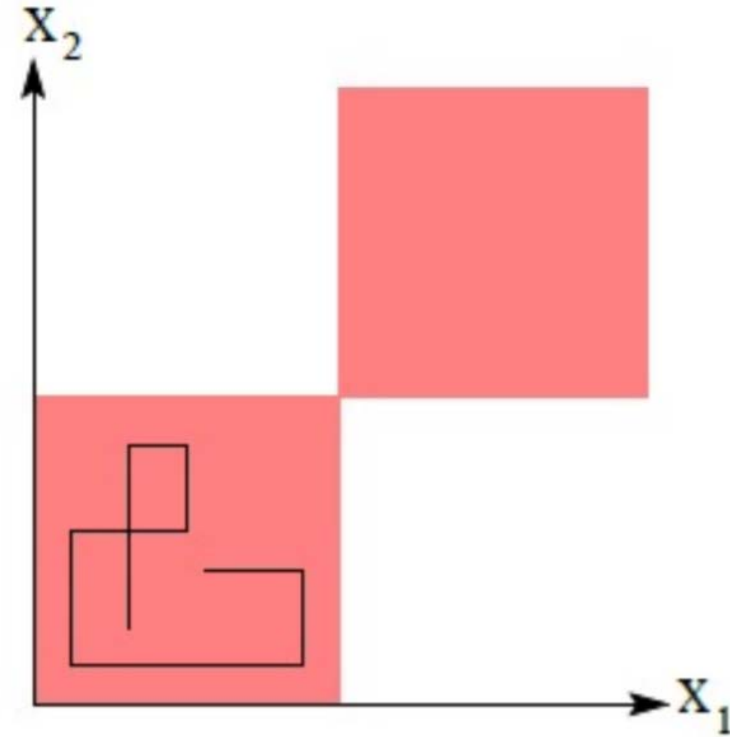
- Will a Gibbs sampler converge to $p(x_1, x_2)$ given the two dimensional distribution on the right?





Markov chain for Gibbs Sampling

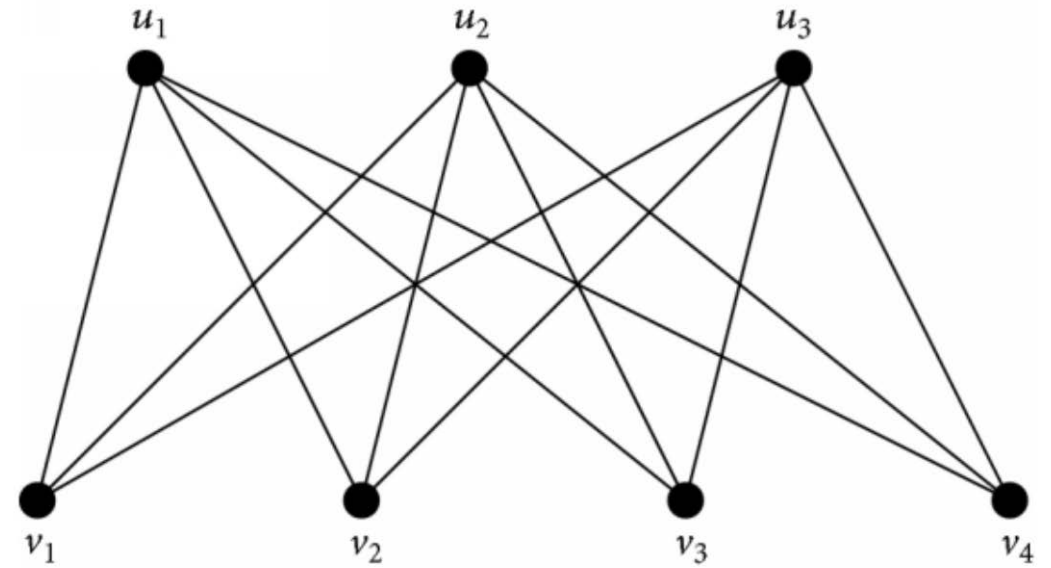
- Will a Gibbs sampler converge to $p(x_1, x_2)$ given the two dimensional distribution on the right?
- No, $p(x|x') = 0$ for x in the upper right quadrant and x' in the lower left (and vice versa).





Markov chain for Gibbs Sampling

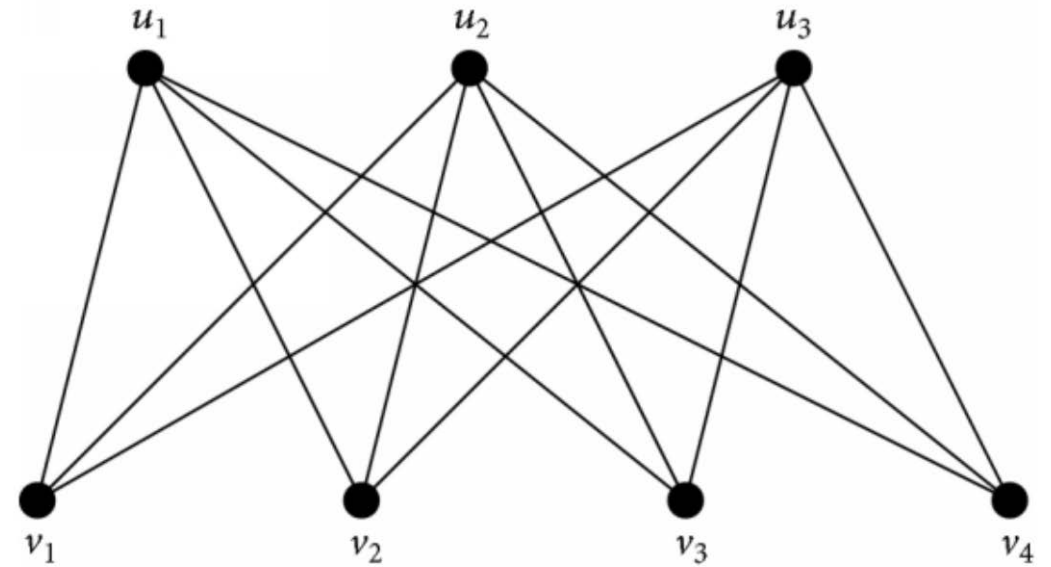
- Will a Gibbs sampler with Markov process defined on the right converge?





Markov chain for Gibbs Sampling

- Will a Gibbs sampler with Markov process defined on the right converge?
- No, bipartite graphs are cyclic with even period



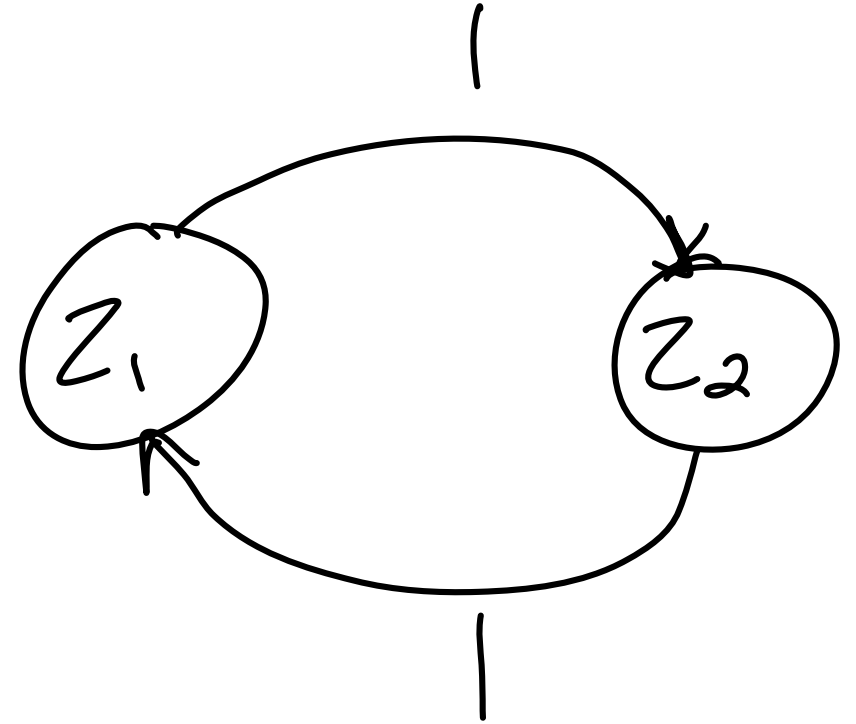


Markov chain for Gibbs Sampling

- Will a Gibbs sampler with Markov process defined on the right converge?
- No, bipartite graphs are cyclic with even period
- Consider transition matrix $T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

And starting points

$(\frac{1}{2}, \frac{1}{2}), (1, 0)$



equations in notes
 $(1, 0)T = (0, 1)$
 $(\frac{1}{2}, \frac{1}{2})T = (\frac{1}{2}, \frac{1}{2})$



Markov chain properties: reversibility

- **Reversible:** an MC is reversible if there exists a distribution $\pi(x)$ such that $\pi(x')T(x | x') = \pi(x)T(x' | x)$

- i.e. chain transitions are symmetric given $\pi(x)$

- **A reversible MC has stationary distribution $\pi(x)$**

- Proof:

$$\pi(x')T(x | x') = \pi(x)T(x' | x) \quad \text{(reversibility)}$$

$$\sum_x \pi(x')T(x | x') = \sum_x \pi(x)T(x' | x)$$

$$\pi(x') \sum_x T(x | x') = \sum_x \pi(x)T(x' | x)$$

$$\pi(x') = \sum_x \pi(x)T(x' | x) \quad \text{(definition of stationary)}$$



Why does MH work?

- We draw $x \sim Q(x'|x)$ and accept according to $A(x'|x)$

- Yields transition kernel $T(x'|x) = Q(x'|x)A(x'|x)$

- The acceptance function is $A(x'|x) = \min \left(1, \frac{P(x')Q(x|x')}{P(x)Q(x'|x)} \right)$

$$\underbrace{A(x'|x)Q(x'|x)P(x)}_{P(x'|x)} = \min \left\{ 1, \frac{P(x')Q(x|x')}{P(x)Q(x'|x)} \right\} Q(x'|x)P(x) \quad \text{acceptance fnc.}$$
$$= \min \{ Q(x'|x)P(x), Q(x|x')P(x') \} \quad \text{multiply through min fnc.}$$
$$= \underbrace{\min \left\{ \frac{Q(x'|x)P(x)}{Q(x|x')P(x')}, 1 \right\}}_{A(x, x')} Q(x|x')P(x') \quad \text{multiply by } \frac{Q(x|x')P(x')}{Q(x|x')P(x')}$$

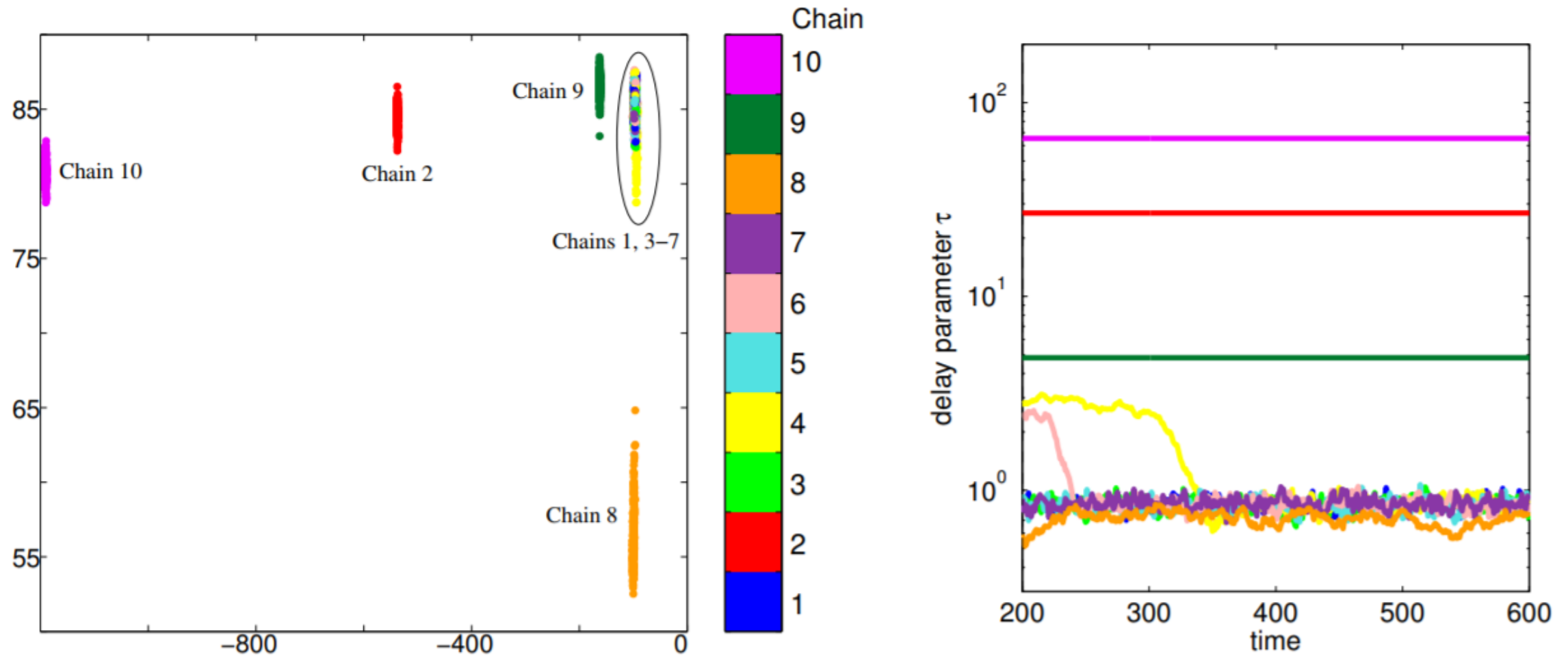
$P(x|x')$ (definition of reversible)

Why does MH work?

- Theorem: If a Markov chain is regular and satisfies reversibility with respect to $p(x)$, then $p(x)$ is its unique stationary distribution
- Easy to verify that Gibbs sampling satisfies aperiodicity and is irreducible, and thus is regular
- The mixing time, or how long it takes to reach something close the stationary distribution, can be very long

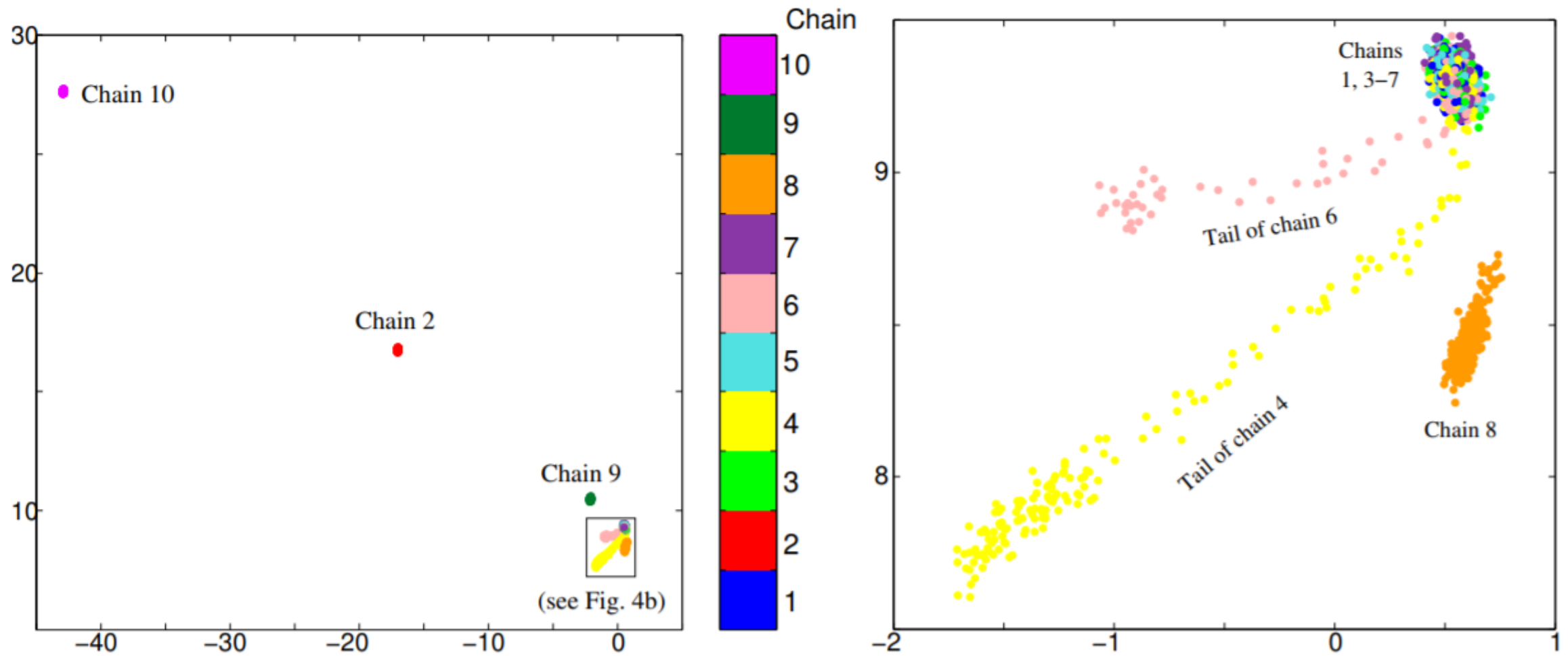


Mixing times: linear discriminant analysis





Mixing time: discriminative component analysis



Variants of MCMC

- Slice Sampling
- Gibbs Sampling
- MH Sampling
- Adaptive MH Sampling
- No-U-Turn Sampling
- Metropolis-adjusted Langevin Algorithm
- Stein Variational Gradient Descent
- Hamiltonian Monte Carlo

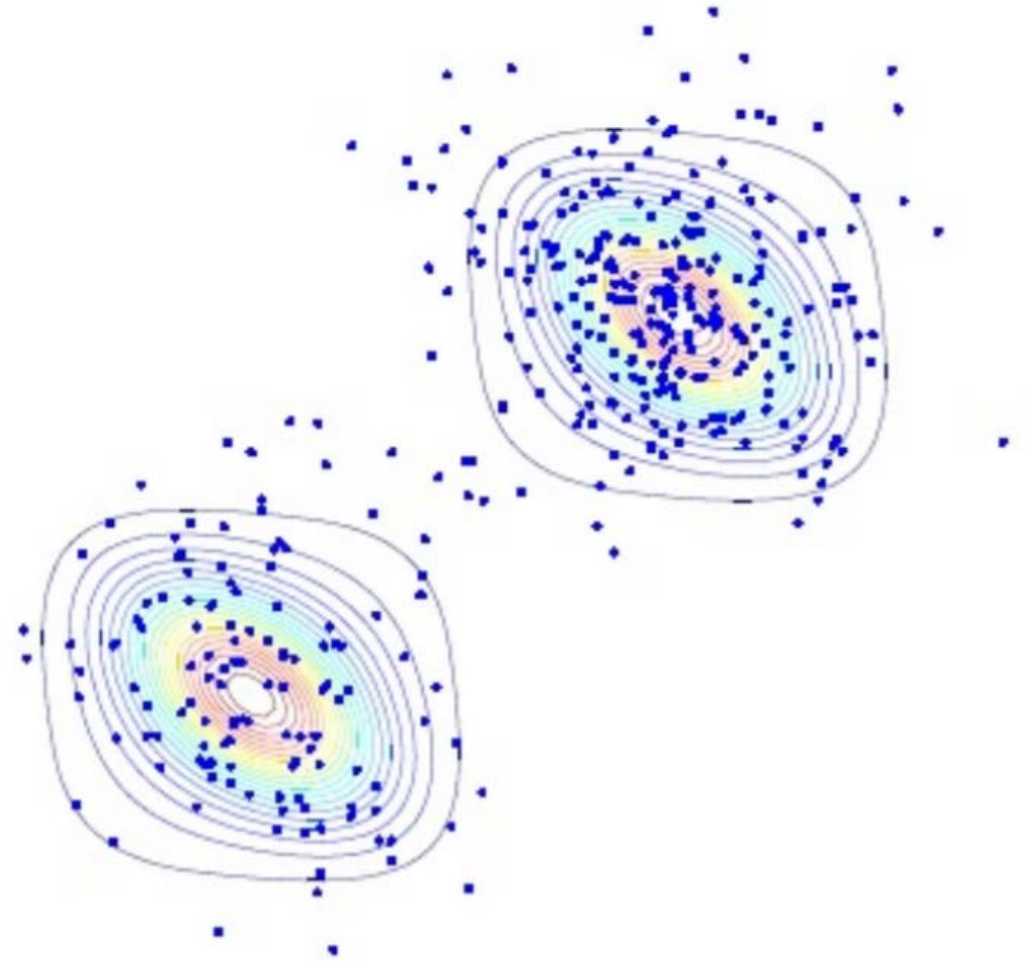
Variants of MCMC

- Slice Sampling
- Gibbs Sampling
- MH Sampling
- Adaptive MH Sampling
- No-U-Turn Sampling
- Metropolis-adjusted Langevin Algorithm
- Stein Variational Gradient Descent
- **Hamiltonian Monte Carlo**



Problems with Gibbs sampling and MH

- Simulations from a multimodal distribution
- Samples from an MH run shown to the right
- Can you explain what is happening?





Hamiltonian Monte-Carlo

- Traditional MCMC proposals based on random walks; slow to mix [[demo](#)]
 - Adding a gradient term removes reversibility
- Hamiltonian MC adds a momentum term to each variable controlling the distance and direction of the jump
- We recover $p(x)$ by forgetting the momentum terms

Hamiltonian Monte-Carlo: a metaphor

- In 2D, a frictionless hockey puck slides over ice surface with varying height
- The state of the system can be described by the position of the puck (2D vector q) and its momentum p (mass $\times$ velocity)
- The potential energy $U(q)$ of the puck is proportional to its height and the kinetic energy $K(p)$ is equal to $\frac{|p|^2}{2m}$
- On a level surface, the puck moves with constant velocity equal to $\frac{p}{m}$
- If it encounters a rising slope, kinetic energy is reduced, potential is gained



Hamiltonian MC

- We write the target distribution as $p(\mathbf{x}) = \frac{1}{Z_x} e^{H_x(\mathbf{x})}$
- We introduce a new auxiliary variable $\mathbf{y}$ $p(\mathbf{y}) = \frac{1}{Z_y} e^{H_y(\mathbf{y})}$
- We assume $\mathbf{x}$ and $\mathbf{y}$ are independent so we can write

$$p(\mathbf{x}, \mathbf{y}) = p(\mathbf{x})p(\mathbf{y}) = \frac{1}{Z_x Z_y} e^{H_x(\mathbf{x}) + H_y(\mathbf{y})} = \frac{1}{Z} e^{H(\mathbf{x}, \mathbf{y})}$$

Hamiltonian MC

- The goal is to generate samples from (x,y) and then throw away y samples after reaching the stationary distribution
- High level algorithm
 - Simulate y' from $p(y)$
 - Simulate x' from $p(x,y')$ using Metropolis-Hastings
- If $H(x', y') \approx H(x, y)$ we will have a high probability of acceptance
- x is interpreted as a position and y is interpreted as a momentum



Hamiltonian MC

- A first order Taylor approximation implies

$$\begin{aligned} H(\mathbf{x}', \mathbf{y}') &= H(\mathbf{x} + \Delta\mathbf{x}, \mathbf{y} + \Delta\mathbf{y}) \\ &\approx H(\mathbf{x}, \mathbf{y}) + \nabla_x H_x(\mathbf{x})^\top \Delta\mathbf{x} + \nabla_y H_y(\mathbf{y})^\top \Delta\mathbf{y} \end{aligned}$$

- In order for this to be a good approximation, we want the last two terms to sum to 0, Hamiltonian MC chooses

$$\Delta\mathbf{x} := \epsilon \nabla_y H(\mathbf{y}) \quad \text{and} \quad \Delta\mathbf{y} := -\epsilon \nabla_x H(\mathbf{x})$$



Hamiltonian MCMC

- Augment model with potential and kinetic energy

Algorithm 27.4 Hybrid Monte Carlo sampling

- 1: Start from $\mathbf{x}^1$
 - 2: **for** $i = 1$ to L **do**
 - 3: Draw a new sample $\mathbf{y}$ from $p(\mathbf{y})$.
 - 4: Choose a random (forwards or backwards) trajectory direction.
 - 5: Starting from $\mathbf{x}^i, \mathbf{y}$, follow Hamiltonian dynamics for a fixed number of steps, giving a candidate $\mathbf{x}', \mathbf{y}'$.
 - 6: Accept the candidate $\mathbf{x}^{i+1} = \mathbf{x}'$ if $H(\mathbf{x}', \mathbf{y}') > H(\mathbf{x}, \mathbf{y})$, otherwise accept it with probability $\exp(H(\mathbf{x}', \mathbf{y}') - H(\mathbf{x}, \mathbf{y}))$.
 - 7: If rejected, we take the sample as $\mathbf{x}^{i+1} = \mathbf{x}^i$.
 - 8: **end for**
-

[\[demo\]](#)

Markov Chain Monte Carlo summary

- General purpose method for Bayesian inference
- Often easy to implement
- Good theoretical guarantees
- Difficult to know when the chain has sufficiently mixed
- Can be very slow to converge
- Many tunable parameters